

Problem 4

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Problem 1 Use the given information to find the equation of the tangent line.

- (a) Given $g(x) = x^3 f(x)$, $f(2) = 4$, and $f'(2) = 7$, find the equation of the tangent line to the graph of g at $x = 2$.

Explanation. Slope of tangent line to the graph of g at the point where $x = 2$:

$$\begin{aligned} g'(x) &= 3x^2 f(x) + x^3 f'(x) \\ \implies g'(2) &= 12(4) + 8(7) = 48 + 56 = 104. \end{aligned}$$

Point on tangent line where $x = 2$: $(2, g(2)) = (2, 8f(2)) = (2, 32)$.

Equation of tangent line:

$$\begin{aligned} y - 32 &= 104(x - 2) \\ \implies y &= 104x - 176. \end{aligned}$$

- (b) Given $h(z) = \frac{zs(z)}{z-3}$, $s(2) = 4$, and $s'(2) = 7$, find the equation of the tangent line to the graph of h at $z = 2$.

Explanation. Slope of tangent line to the graph of h at the point where $z = 2$:

$$\begin{aligned} h'(z) &= \frac{(z-3)(zs(z))' - zs(z)(z-3)'}{(z-3)^2} \\ &= \frac{(z-3)(s(z) + zs'(z)) - zs(z)(1)}{(z-3)^2} \\ \implies h'(2) &= \frac{(2-3)(s(2) + 2s'(2)) - 2s(2)}{(2-3)^2} = -26. \end{aligned}$$

Point on tangent line where $z = 2$: $(2, h(2)) = (2, \frac{2s(2)}{2-3}) = (2, -8)$.

Equation of tangent line to the graph of h at the point where $z = 2$:

$$\begin{aligned} y - (-8) &= -26(z - 2) \\ \implies y &= -26z + 44. \end{aligned}$$

(c) Given

x	1	2	3	4	5
$f(x)$	5	3	0	-4	3
$f'(x)$	-3	-5	-2	6	-4
$g(x)$	6	9	-8	13	15
$g'(x)$	8	5	-10	7	6

find the equation of the tangent line of

$$\frac{f(x)}{e^x g(x)}$$

at $x = 2$.

Explanation. Slope of tangent line to the graph of $\frac{f(x)}{e^x g(x)}$ at the point where $x = 2$:

$$\begin{aligned} \left(\frac{f(x)}{e^x g(x)} \right)' &= \frac{e^x g(x) f'(x) - f(x)(e^x g(x) + e^x g'(x))}{(e^x g(x))^2} \\ \implies \frac{d}{dx} \left(\frac{f(x)}{e^x g(x)} \right) \Big|_{x=2} &= \frac{e^2 g(2) f'(2) - f(2)(e^2 g(2) + e^2 g'(2))}{(e^2 g(2))^2} \\ &= \frac{e^2(9)(-5) - (3)(9e^2 + 5e^2)}{(9e^2)^2} \\ &= \frac{-45e^2 - 42e^2}{81e^4} \\ &= \frac{-87e^2}{81e^4} \\ &= \frac{-87}{81e^2} \end{aligned}$$

Point on tangent line where $x = 2$: $(2, f(2)/(e^2 g(2))) = (2, 3/(9e^2)) = (2, 1/(3e^2))$.

Equation of tangent line where $x = 2$:

$$\begin{aligned} y - \frac{1}{3e^2} &= \frac{-87}{81e^2}(x - 2) \\ \implies y &= \frac{-87}{81e^2}x + \frac{67}{27e^2}. \end{aligned}$$